

Python vs R

Task	Python (pandas)	R (dplyr or base R)
Load Data	<code>df = pd.read_csv("file.csv")</code>	<code>df <- read.csv("file.csv")</code>
Filter Rows	<code>df[df['column'] > 100]</code>	<code>filter(df, column > 100)</code> or <code>df[df\$column > 100,]</code>
Select Columns	<code>df[['col1', 'col2']]</code>	<code>select(df, col1, col2)</code> or <code>df[, c("col1", "col2")]</code>
Sort Data	<code>df.sort_values(by='column', ascending=False)</code>	<code>arrange(df, desc(column))</code> or <code>df[order(-df\$column),]</code>
Group By / Aggregate	<code>df.groupby('dept').agg({'salary': 'mean'})</code>	<code>df %>% group_by(dept) %>% summarise(mean_salary = mean(salary))</code>
Count Rows	<code>len(df)</code> or <code>df.shape[0]</code>	<code>nrow(df)</code>
Average / Mean	<code>df['salary'].mean()</code>	<code>mean(df\$salary)</code>
SUM	<code>df['sales'].sum()</code>	<code>sum(df\$sales)</code>
Remove Duplicates	<code>df.drop_duplicates()</code>	<code>distinct(df)</code> or <code>unique(df)</code>
Join Tables	<code>pd.merge(df1, df2, on='id')</code>	<code>left_join(df1, df2, by="id")</code>
Create New Column	<code>df['bonus'] = df['salary'] * 0.1</code>	<code>df <- mutate(df, bonus = salary * 0.1)</code>
Rename Column	<code>df.rename(columns={'old': 'new'}, inplace=True)</code>	<code>rename(df, new = old)</code> or <code>names(df)[names(df)=="old"] <- "new"</code>
Handle Missing Data	<code>df.fillna(0)</code> or <code>df.dropna()</code>	<code>df[is.na(df)] <- 0</code> or <code>na.omit(df)</code>
Export Data	<code>df.to_csv("file.csv", index=False)</code>	<code>write.csv(df, "file.csv", row.names = FALSE)</code>
Data Visualization	<code>df.plot()</code> or <code>seaborn</code> , <code>matplotlib</code>	<code>plot(df)</code> or use <code>ggplot2</code>